

CMSC3621 Data Structures and Algorithms Lab

Lab Assignment 7

Overview

This assignment is for the students to review and practice hashing with chaining and implementing a simple hash table.

Details

In the attached .zip file, we provide a C++ project.

You are to follow the comments of the functions to complete the code of `insert()`, `search()` and `remove()`. When you finish the `insert()`, `search()` and `remove()` functions, please run the code in `main()`, but do not change the code in `main.cpp`.

The STL container `deque` is used to simulate a liner data structure. If you are not familiar with `deque`, you can refer to the attached slide which is an overview of STL including the details of `deque`.

Submission

- 1) `HashTable.h` //Comments are needed to illustrate how do you implement the code
- 2) A report (a PDF file), in the report you only need to provide screenshots and **descriptions** about how do you test your implementation and why your solution is correct according to the output

Important requirements

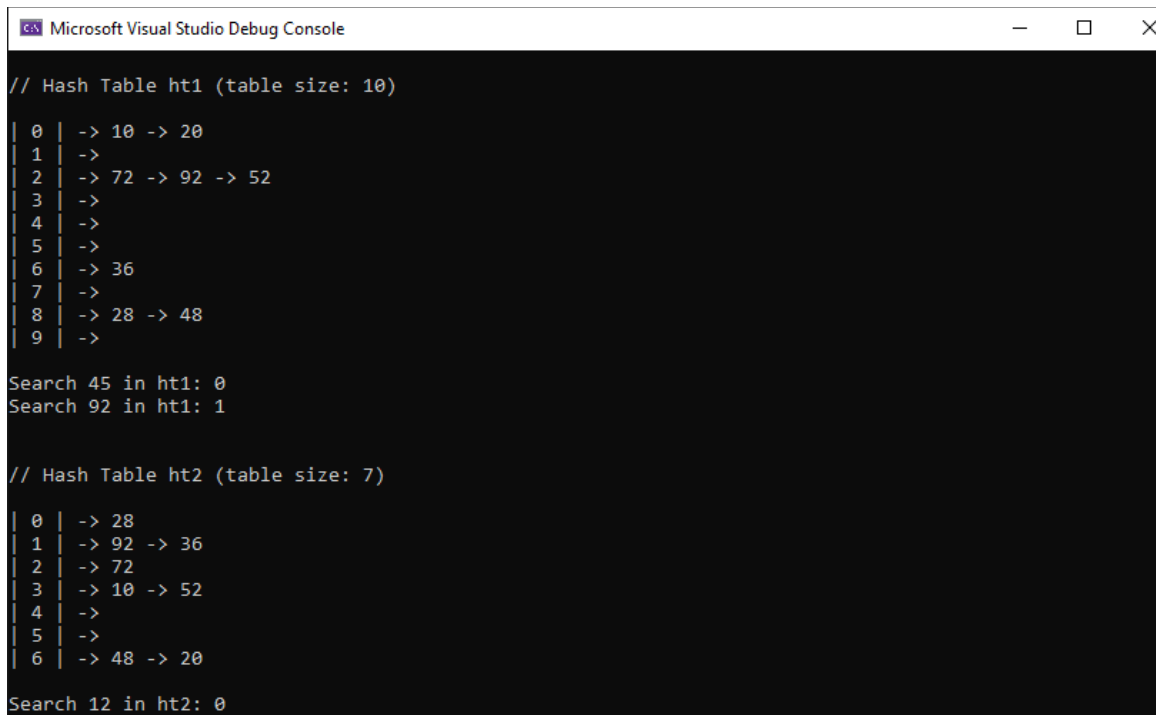
- 1) Don't include any other files (that are not aforementioned).
- 2) Don't modify anything in the existing files / project, otherwise your code may not be able to run in the grader's environment.
- 3) You only can write your code at certain places which are indicated by the existing comments.

4) Any violation against the rules above will lead to extra penalty. If you missed the report, your submission will not be evaluated.

5) The following screenshot shows the correct output of the program for the first 2 hash tables. You must provide all outputs of all hash tables defined in the main().

5.1) If you only attach the first two outputs like below or just reuse the screenshot provided below, you will get zero for this assignment.

5.2) You must run your code in Visual Studio. As a result, the top-left corner of your command line window should display 'Microsoft Visual Studio ...' (as below). If this item is not followed, a 10 points penalty will be applied.



```
Microsoft Visual Studio Debug Console

// Hash Table ht1 (table size: 10)

| 0 | -> 10 -> 20
| 1 | ->
| 2 | -> 72 -> 92 -> 52
| 3 | ->
| 4 | ->
| 5 | ->
| 6 | -> 36
| 7 | ->
| 8 | -> 28 -> 48
| 9 | ->

Search 45 in ht1: 0
Search 92 in ht1: 1

// Hash Table ht2 (table size: 7)

| 0 | -> 28
| 1 | -> 92 -> 36
| 2 | -> 72
| 3 | -> 10 -> 52
| 4 | ->
| 5 | ->
| 6 | -> 48 -> 20

Search 12 in ht2: 0
```

Rubric

1) There are three incomplete functions in total. Partial credits may be given in terms of to what degree you can complete them.

1.1) insert(): 25 pts

1.2) search(): 24 pts

1.3) remove(): 30 pts

2) Report: 20 pts